

Digital Egypt Pioneers Initiative - Data Analysis Track

Enterprise E-Commerce Intelligence & Predictive Analytics

GRADUATION PROJECT DOCUMENTATION

Team Members

Duaa Abd-Elati

Hazem Mahmoud

Mai Gamal

Maya Mohamed

Osama Hussein

Ahmed Ezzat

Supervisor: Eng. Mohamed Alemam

Mentor: Eng. Mahmoud

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Chapter 1

Project Overview

Chapter 1: Project Overview

1.1 Project Concept

The rapid transformation of global digital commerce generates massive volumes of transactional data, yet organizations frequently struggle to translate granular operations into executive decision-making. This graduation project, developed under the **Digital Egypt Pioneers Initiative (DEPI)**, implements an end-to-end Modern Data Stack to bridge the gap between complex transactional records and actionable business strategy.

By transitioning raw e-commerce data into a centralized, structured Data Warehouse following the **Medallion Architecture**, the project establishes a Single Source of Truth (SSOT). This foundation enables the deployment of advanced machine learning models to automate demand forecasting and actively detect margin-eroding products ("Bleeding SKUs"). The data engineering and analytics workflow culminates in a multi-page interactive **Power BI** dashboard suite designed to drive real-time executive decisions across four analytical pillars:

- **Product Intelligence & Portfolio Optimization:** Segmenting high-volume "Hero" products from high-margin "Niche" products, utilizing Association Rule Learning (Apriori Algorithm) for bundling strategies, and flagging margin erosion.
- **Machine Learning & Predictive Analytics:** Anticipating unit sales over a 60-to-90-day horizon, modeling price elasticity, and automating anomaly detection for refund spikes or conversion drops.
- **Advertising (PPC) & Halo Effect Analysis:** Benchmarking Total ACOS (TACoS) to evaluate organic profitability and statistically quantifying the long-term organic ranking lift generated by paid advertising spend.
- **Financial Risk & Operational Efficiency:** Deconstructing net profit margins to isolate logistics costs (FBA fees), promotional cannibalization, and marketplace variances across global regions.

1.2 Data Source

The foundational dataset comprises over 13,000 granular, real-world corporate transaction records sourced from a global e-commerce corporation operating across multiple European marketplaces.

Attribute	Detail
Dataset Domain	Global E-Commerce Operations & Transactional Records
Volume	Over 13,000 detailed enterprise transaction records
Geographic Coverage	Multi-Marketplace Europe (UK, Germany [DE], France [FR], Italy [IT], Spain [ES])
Key Metrics Tracked	SKU identifiers, PPC Advertising Spend, Fulfillment by Amazon (FBA) Fees, COGS, Promotional Discounts, Refunds, Sessions, Clicks, and Page Views
Processing Stack	Relational SQL Database, Python (Pandas/Scikit-Learn/Prophet), Excel, and Power BI
File format	Excel (.xlsx)
Rows used in project	Full downloaded records

1.3 Dataset Structure

The dataset comprises 37 attributes describing sales transactions, product information, financial performance, customer traffic, advertising activities, and profitability metrics across multiple Amazon marketplaces.

The following table summarizes the main dataset columns, their data types, and their business purpose.

Column Name	Data Type	Description
Date	Date	Transaction date.
Title	Text	Product title displayed on Amazon.
Account Title	Text	Seller account associated with the transaction.
Market Place	Text	Marketplace where the sale occurred (UK, DE, FR, IT, ES).
SKU	Text	Unique Stock Keeping Unit identifier.
FNSKU	Text	Fulfillment Network SKU assigned by Amazon.
ASIN	Text	Amazon Standard Identification Number.
Parent ASIN	Text	Parent product identifier for product variations.
Is Parent	Boolean	Indicates whether the product is a parent listing.
Internal Name	Text	Internal product name used by the company.
Brand	Text	Product brand name.
Product Group	Text	Product category or group.
Taxes	Decimal	Taxes deducted from the transaction.
Orders	Whole Number	Number of customer orders.
Units	Whole Number	Number of units sold.
Refunded	Whole Number	Number of refunded units.
Refund %	Percentage	Percentage of refunded units.

Unit Session %	Percentage	Conversion rate from sessions to purchased units.
Promo Units	Whole Number	Units sold during promotional campaigns.
Organic Units	Whole Number	Units sold without paid advertising.
Per Unit Revenue	Decimal	Average revenue generated per unit sold.
Revenue	Currency	Total sales revenue.
COGS	Currency	Cost of Goods Sold.
FBA Fees	Currency	Amazon fulfillment service fees.
Promo Amount	Currency	Total promotional discount amount.
Sessions	Whole Number	Number of customer sessions.
Page Views	Whole Number	Number of product page views.
Impressions	Whole Number	Number of advertisement impressions.
Clicks	Whole Number	Number of advertisement clicks.
PPC Orders	Whole Number	Orders generated from PPC campaigns.
PPC Sales	Currency	Sales generated through paid advertising.
PPC Cost	Currency	Advertising cost spent on PPC campaigns.
PPC Conv	Percentage	PPC conversion rate.
OOE	Decimal	Operational efficiency metric used by the business.
Net Profit	Currency	Final profit after deducting all operational costs.
Net Margin	Percentage	Net profit expressed as a percentage of revenue.
Net ROI	Percentage	Return on Investment after considering all expenses.

The dataset combines transactional, financial, operational, and marketing information, allowing comprehensive analysis of product performance, profitability, customer behavior, advertising effectiveness, and business growth across multiple marketplaces.

1.4 Project Impact (Key Performance Indicators)

To evaluate brand health and operational efficiency beyond raw revenue, the project derives a suite of core executive KPIs:

Dashboard 1 – Executive Overview

KPI	Definition	Calculation
Total Revenue	Total sales generated across all marketplaces.	SUM(Revenue)
Total Net Profit	Final profit after deducting all operational costs.	SUM(Net Profit)
Total Units Sold	Total number of units sold.	SUM(Units)
Total Orders	Total customer orders.	SUM(Orders)
Net ROI	Return on investment after all expenses.	Net Profit ÷ Total Cost
Total PPC Cost	Total advertising spend.	SUM(PPC Cost)

Dashboard 2 – SKU Performance

KPI	Definition	Calculation
Total Distinct Products	Number of unique SKUs.	DISTINCTCOUNT(SKU)
Top Product Revenue	Highest revenue generated by a single SKU.	MAXX(VALUES(SKU), [Revenue])
Revenue per Product	Average revenue generated per SKU.	Revenue ÷ Distinct Products
Total COGS	Total Cost of Goods Sold.	SUM(COGS)
Bleeding SKUs	Number of products with negative profit.	COUNTROWS(FILTER(SKU, Net Profit < 0))

Dashboard 3 – Marketplace Performance

KPI	Definition	Calculation
Revenue per Unit	Average revenue generated from each sold unit.	Revenue ÷ Units
Total Orders	Total orders across marketplaces.	SUM(Orders)
Average Net Margin	Average profitability percentage across marketplaces.	AVERAGE(Net Margin)
Average Net ROI	Average ROI across marketplaces.	AVERAGE(Net ROI)

Dashboard 4 – Marketing & Traffic (Halo Effect)

KPI	Definition	Calculation
Total Sessions	Total customer sessions.	SUM(Sessions)
Average CTR	Average Click Through Rate.	AVERAGE(CTR)
Total PPC Spend	Total advertising cost.	SUM(PPC Cost)
Total PPC Sales	Revenue generated from paid ads.	SUM(PPC Sales)
Average Conversion Rate	Average visitor conversion rate.	AVERAGE(Unit Session %)

Dashboard 5 – Refund Leakage & Quality Control

KPI	Definition	Calculation
Total Refunded Units	Total units returned by customers.	SUM(Refunded)
Refund Rate	Percentage of refunded units.	Refunded ÷ Units
High-Risk SKUs	Number of SKUs exceeding refund threshold.	COUNTROWS(FILTER(SKU, Refund % > 5%)) <i>(or your chosen threshold)</i>
Revenue Lost Due to Refunds	Estimated revenue lost because of refunds.	SUM(Refund Amount) <i>(or Refunded × Per Unit Revenue)</i>
Highest Refund Marketplace	Marketplace with the highest refund amount.	Top Marketplace by Refund Amount

Dashboard 6 – Promotions Analysis

KPI	Definition	Calculation
Total Promo Units	Units sold through promotions.	SUM(Promo Units)
Total Promo Amount	Total promotional discounts given.	SUM(Promo Amount)
Promo ROI	Return generated from promotional campaigns.	Promo Revenue ÷ Promo Amount
Promo Conversion Rate	Percentage of sessions converted during promotions.	Promo Orders ÷ Promo Sessions <i>(or your existing Promo Conversion measure)</i>

Dashboard 7 – AI & Machine Learning

KPI	Definition	Calculation
Model Accuracy	Accuracy achieved by the Random Forest model.	$(\text{Correct Predictions} \div \text{Total Predictions}) \times 100$
Precision	Percentage of predicted positive SKUs that were actually bleeding.	$TP \div (TP + FP)$
Recall	Percentage of actual bleeding SKUs correctly detected.	$TP \div (TP + FN)$
F1 Score	Harmonic mean of Precision and Recall.	$2 \times (\text{Precision} \times \text{Recall}) \div (\text{Precision} + \text{Recall})$
Predicted Bleeding SKUs	Number of products predicted as high risk.	COUNTROWS(Predicted Bleeding SKUs)

KPIs for Machine Learning Dashboard:

1.4 Main Charts Overview

The Power BI suite spans six distinct, purpose-built command centers:

Page	Title	Focus Area	Key Visualizations
1	Executive Command & Overall Performance	Top-level financial health and profitability velocity	True Profitability Combo Chart, Cost Structure Waterfall, TACoS vs. Net Margin Dual Line
2	Product Portfolio & "Bleeding SKUs"	SKU-level margin analysis and unprofitable item identification	Pareto 80/20 Profit Chart, Vanity vs. Sanity Matrix, Bleeding SKUs Alert Table
3	Marketplace & Geographic Performance	Regional expansion, localized ROI, and marketplace fee burden	Revenue vs. Net ROI Bar Chart, Day-of-Week Profitability Heatmap, Marketplace Cost Breakdown
4	Marketing, Traffic & "The Halo Effect"	PPC optimization, traffic quality, and organic search ranking lift	Traffic vs. Conversion Scatter Plot, Halo Effect Tracker, PPC Funnel Health
5	Refund Leakage & Quality Control	Systemic cash bleed mitigation from returns and listing defects	Financial Impact Area Chart, Refund Worst Offenders Bar Chart, Promo-Induced Returns Comparison
6	Promotion Cannibalization & Velocity	Discount effectiveness, customer acquisition vs. cannibalization	Revenue Composition Stacked Bar, Cannibalization Tracker, Price Elasticity Matrix

1.5 SMART Research Questions

The analytical workflow addresses eight SMART (Specific, Measurable, Achievable, Relevant, and Time-bound) business questions to support strategic decision-making:

Metric Category	Key Performance Indicator	Definition & Mathematical Formulation	Target Benchmark
Financial Health	Net Profit Margin	True bottom-line profitability after accounting for COGS, FBA fees, PPC costs, and refunds. $Margin = \frac{Total\ Net\ Profit}{Total\ Revenue}$	> 15%
Advertising Efficiency	Total ACOS (TACoS)	The true cost of advertising relative to combined organic and paid revenue. $TACoS = \frac{\sum PPC\ Cost}{\sum Total\ Sales}$	< 10% (for mature brands)
Inventory Management	Inventory Velocity	Average units sold per day to optimize reorder points and prevent storage fee accumulation.	Dynamic SKU target
Quality Control	Refund Rate %	Proportion of units returned relative to total units shipped.	< 3%
Predictive Performance	Model MAPE	Mean Absolute Percentage Error measuring forecasting model precision.	< 10%

1. Executive Performance: How do Revenue, Net Profit, and Net ROI change over time, and what are the main factors affecting overall business profitability?
2. SKU Performance: Which SKUs generate the highest profit, and which products are classified as Bleeding SKUs due to low margins or negative profitability?
3. Marketplace Performance: Which marketplace (UK, DE, FR, IT, or ES) achieves the highest Revenue and ROI, and which incurs the highest operational costs?
4. Marketing & Halo Effect: Does increasing PPC advertising spend improve organic sales performance, and how effective are marketing campaigns in driving conversions?
5. Refund & Quality Control: Which products and marketplaces have the highest refund rates, and how do refunds impact overall financial performance?
6. Promotions Analysis: Do promotional campaigns increase sales and customer acquisition while maintaining acceptable profitability and ROI?
7. Machine Learning Prediction: Can the Random Forest model accurately identify high-risk Bleeding SKUs before they negatively impact profitability?
8. Business Decision Support: How can interactive Power BI dashboards help executives optimize pricing, advertising, inventory, and operational performance through data-driven insights?

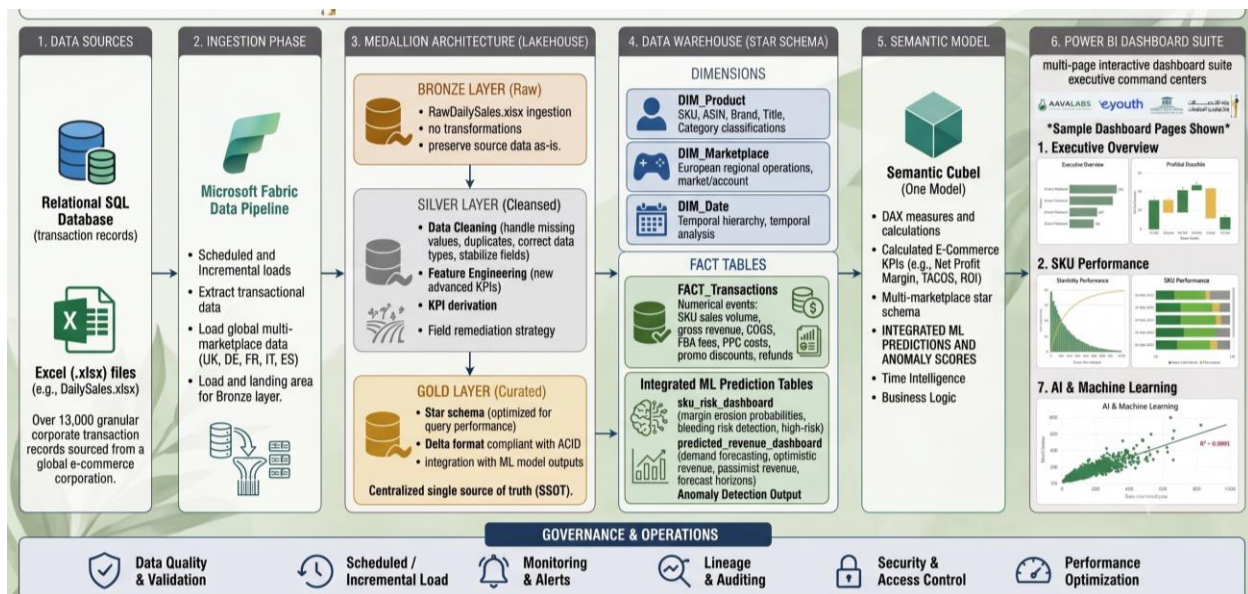
Chapter 2

Technical Architecture

Chapter 2: Technical Architecture

2.1 Technical Architecture

This chapter describes the technologies, architectural design, and analytical workflow used to transform raw e-commerce data into interactive dashboards and predictive insights. The project combines Python for data processing and machine learning, Power BI for business intelligence and dashboard development, and Figma for UI/UX design and dashboard prototyping.



2.2 Tools Used

The project was developed using four primary technologies covering data engineering, data processing, machine learning, dashboard development, and user interface design.

Tool / Platform	Role in Project	Key Usage
Microsoft Fabric	Data Engineering & Data Storage Platform	Used to organize the project workflow using the Medallion Architecture (Bronze, Silver, and Gold layers), manage datasets, perform data transformation, and prepare structured data for analytics and reporting.
Python	Data Processing & Machine Learning	Used for data cleaning, preprocessing, feature engineering, KPI calculations, data analysis, and developing the Random Forest model for SKU risk prediction before exporting the processed datasets.
Power BI	Business Intelligence & Interactive Dashboard	Used to build seven interactive dashboard pages, create DAX measures, KPI cards, charts, slicers, drill-through navigation, bookmarks, and executive-level business reports.
Figma	UI/UX Design & Dashboard Prototyping	Used to design the dashboard layout, navigation sidebar, color palette, typography, page structure, and overall visual hierarchy before implementation in Power BI.

2.2 Architecture (Medallion)

The Medallion Architecture organizes data into three logical layers - Bronze, Silver, and Gold - each serving a specific purpose. This architecture ensures raw data is never overwritten (Bronze), all cleaning logic is centralized (Silver), and the analytical layer (Gold) is purpose-built for Power BI performance.

➤ Medallion Layers

Bronze Layer	Silver Layer	Gold Layer
Raw ingestion, Excel	Data cleaning	Star schema
No transformation	Feature engineering	3 dim + 1 fact
Files/Bronze/	KPI derivation Files/Silver/	Delta format Tables/Gold/

2.3 Data Model (Star Schema)

The Gold layer implements a star schema consisting of one central Fact Table surrounded by five Dimension Tables, each connected via the User_ID surrogate key. All tables are saved in Delta format for ACID compliance and direct Power BI connectivity.

Tables Description (Data Schema)

Table	Type	Columns / Keys
fact	Fact	Date_ID, Market_ID, SKU_ID, Clicks, COGS, FBA Fees, Impressions, Net Margin, Net Profit, Net ROI, OOE, Orders, Organic Units, Page Views, Per Unit Revenue, PPC Conv, PPC Cost, PPC Orders, PPC Sales, Promo Amount, Promo Units, Refund %, Refunded, Revenue, Sessions, Taxes, Units
Dim_date	Dimension	Date_ID, Date, Year, Quarter, Month, Day
Dim_market_account	Dimension	Market_ID, Account Title, Market Place
Dim_product	Dimension	SKU_ID, ASIN, Brand, FNSKU, Is Parent, Parent ASIN, SKU, Title

2.4 Cleaning Process Across Bronze & Silver Layers

2.4.1 Bronze layer:

The Bronze Layer represents the raw ingestion stage of the Medallion Architecture, where the original dataset is imported and stored without applying any transformations. The purpose of this layer is to preserve the source data in its original format and perform an initial assessment of its structure and quality before any cleaning or preprocessing.

The following steps were performed in the Bronze Layer:

- Imported the raw **DailySales.xlsx** dataset into Python using the **Pandas** library.
- Displayed the first few records using **head()** to verify that the data was loaded correctly.

- Inspected the dataset structure using **info()** to identify data types, missing values, and column information.
- Checked the dataset dimensions using **shape()**, confirming that the dataset contains **13,324 rows** and **37 columns**.
- Retrieved the complete list of column names using **columns** for schema validation.
- Exported the dataset as **raw_data.csv**, preserving the original data without any modifications to serve as the input for the Silver Layer.

2.4.2 Silver layer

The Silver Layer is responsible for transforming the raw data stored in the Bronze Layer into a clean, consistent, and analysis-ready dataset. During this stage, data quality issues were resolved through a comprehensive cleaning process, including handling missing values, removing duplicate records, correcting data types, and standardizing text fields. Invalid records and inconsistencies were also identified and treated to improve data reliability.

In addition to data cleaning, feature engineering techniques were applied to enrich the dataset and prepare it for business analysis and machine learning. New business-related features were derived from the cleaned transactional data, enabling more meaningful analysis and supporting advanced reporting in the Gold Layer.

The output of the Silver Layer is a structured and validated dataset that serves as the foundation for dashboard development, KPI calculation, and predictive analytics.

Field Name	Detected Anomaly	Remediation Strategy
PPC_Cost	Missing/Null records on organic sales	Imputed with 0.00 to reflect zero advertising expenditure
Promo_Amount	Unrecorded discount fields	Imputed with 0.00 to standardize full-price unit sales
Transaction_ID	Duplicate entry attempts	deduplication preserving earliest timestamp
Refund_Amount	Inconsistent sign entry ($\pm$)	Standardized to absolute positive values for waterfall accounting

2.3 Data Model (Star Schema)

The Gold Layer structures historical and machine learning datasets into a clean star schema to ensure optimal query performance in Power BI:

- **Fact Table (fact_transactions):** Stores granular numerical events including SKU sales volume, gross revenue, COGS, FBA fees, PPC costs, promo discounts, and refund amounts.
- **Dimension Tables:**
 - dim_date: Temporal hierarchy (Year, Quarter, Month, Week, Day of Week) supporting time-series forecasting and seasonality mapping.
 - dim_product: SKU metadata, product titles, category classifications, and baseline cost benchmarks.
 - dim_marketplace: Geographic data covering UK, DE, FR, IT, and ES regional operations.
- **Integrated ML Prediction Tables:**
 - predicted_revenue_dashboard: Contains daily forecast horizons (Predicted_Revenue, Pessimistic_Revenue, Optimistic_Revenue) generated by Facebook Prophet.
 - sku_risk_dashboard: Contains SKU-level margin erosion probabilities (Predicted_Bleeding_Risk) generated by the Random Forest Classifier.

<https://github.com/DuaA-A/Enterprise-E-Commerce-Predictive-Analytics/tree/main/Medallion%20Architecture>

2.4 Cloud Modernization via Microsoft Fabric

To elevate the project to an enterprise-grade cloud environment, the data pipeline and analytics workflows were integrated into **Microsoft Fabric**, an all-in-one analytics solution. This integration modernizes our Medallion Architecture by unifying Data Engineering, Data Science, and Business Intelligence into a single SaaS platform.

- **Unified Storage (OneLake):** Instead of relying on isolated local files and CSVs, the Bronze, Silver, and Gold layers are migrated to Microsoft OneLake. This provides a single, governed source of truth across the entire organization, eliminating data silos and reducing data movement overhead.
- **Fabric Data Engineering:** The data cleaning, smart imputation, and schema modeling previously executed in isolated Jupyter Notebooks are transitioned into **Fabric Notebooks** and **Data Factory Pipelines**. This allows the ETL (Extract, Transform, Load) processes to run on powerful, distributed Apache Spark compute clusters, ensuring the pipeline can effortlessly scale to handle millions of rows of future e-commerce transactions.
- **Direct-to-Power BI Integration:** By leveraging the Gold Layer natively within Fabric, the Power BI dashboards connect via DirectLake mode. This eliminates the need for manual data imports or scheduled refreshes, ensuring that executives and stakeholders are viewing real-time inventory levels, refund metrics, and sales forecasting without query latency.

By implementing Microsoft Fabric, the project achieves true end-to-end automation, reducing infrastructure complexity while vastly improving data governance, security, and computational scalability.

Chapter 3

Data Engineering & Analysis

Chapter 3: Data Engineering & Analysis

To align raw statistical findings with narrative-driven business insights, the EDA environment is uniquely structured to feed directly into six specialized intelligence dashboards:

3.1 Executive Overview (Dashboard 1)

- **File Reference:** Dashboard1_Overview.ipynb
- **Focus:** High-level key performance indicators (KPIs) and macroscopic business health.
- **Analysis:** Aggregation of total net revenue, overall order volume, and top-line growth trends over time. This phase establishes the baseline financial metrics against which detailed, granular data is measured.

3.2 Deep-Dive Performance Profiling (Dashboards 2 & 3)

- **File Reference:** Dashboard2_EDA.ipynb, Dashboard3_EDA.ipynb
- **Focus:** SKU-level performance matrices, temporal seasonality, and categorical dominance.
- **Analysis:** Identifies the "Star Performers" and "Laggards" within the product portfolio. Time-series analysis is applied to uncover seasonal purchasing patterns and peak sales velocity periods, which is critical for precise inventory and supply chain planning.

3.3 Market & Traffic Analytics (Dashboard 4)

- **File Reference:** EDA_For_Dashboard_4.ipynb
- **Focus:** User acquisition, conversion tracking, and advertising efficiency.
- **Analysis:** Explores the relationship between page views, user sessions, and final transaction conversions. This analysis pinpoints highly-trafficked items that suffer from low conversion rates, highlighting immediate opportunities for listing content optimization and PPC adjustments.

3.4 Refund Leakage & Quality Control (Dashboard 5)

- **File Reference:** Dashboard5_Refund_Leakage_&_Quality_Control.ipynb
- **Focus:** Isolating operational defects, reverse logistics impacts, and revenue attrition.
- **Analysis:** Aggregates refund frequencies at the Parent ASIN and individual SKU levels. By mapping refund metrics against total sales volume, this notebook identifies "Bleeding SKUs"—products that generate high initial top-line revenue but suffer from margin-destroying return rates due to potential manufacturing or description defects.

3.5 Promotions & Returns Analysis (Dashboard 6)

- **File Reference:** Dashboard6_Promotions_&_Returns_Analysis.ipynb
- **Focus:** Evaluating the true ROI of discounting strategies and promotional campaigns.
- **Analysis:** Quantifies the financial impact of active promotions. It investigates "Promo Cannibalization"—scenarios where aggressive discounts inadvertently subsidize organic sales rather than acquiring net-new customers, thereby diminishing the bottom-line net profit.

3.6 Outcomes of the EDA Phase

The analytical rigor applied in this chapter ensures that the subsequent Business Intelligence layer is not merely displaying raw data, but visualizing pre-validated, high-impact business realities. By isolating operational anomalies (e.g., refund leakage) and calculating true net margins, the EDA phase provides the essential statistical backing required for the Predictive Machine Learning models and executive decision-making processes discussed in subsequent chapters.

<https://github.com/DuaA-A/Enterprise-E-Commerce-Predictive-Analytics/tree/main/Analysis/EDA>

Chapter 4

Machine Learning & Predictive Analytics

4.1 Introduction

Traditional Business Intelligence focuses on analyzing historical data to explain what has already happened. However, modern e-commerce organizations require intelligent systems capable of identifying unusual business behavior automatically.

To enhance the analytical capabilities of the project, an Artificial Intelligence module was developed using Machine Learning techniques in Python. The objective of this module is to detect abnormal sales patterns, unexpected revenue fluctuations, and unusual operational behavior that may indicate potential business risks.

This predictive approach enables proactive decision-making by identifying anomalies before they significantly impact business performance.

4.2 Machine Learning Workflow

The Machine Learning pipeline follows a structured workflow that transforms prepared business data into predictive insights.

The workflow consists of the following stages:

1. Loading the prepared Gold Layer dataset.
2. Data preprocessing and feature selection.
3. Preparing numerical features for model training.
4. Training the anomaly detection model.
5. Predicting anomalies for each transaction.
6. Evaluating the prediction results.
7. Exporting the prediction output for business reporting.

This workflow ensures that the anomaly detection process is reliable, scalable, and suitable for integration with business intelligence solutions.

4.3 Data Preparation

Before training the machine learning model, the dataset was prepared to improve prediction quality and ensure consistent model performance.

The preprocessing stage included:

- Reviewing the cleaned Gold Layer dataset.
- Handling missing and invalid values.
- Selecting relevant numerical features.
- Removing unnecessary categorical attributes.
- Preparing the dataset for unsupervised learning.

These preprocessing steps improve the accuracy and stability of the anomaly detection model.

4.4 Selected Features

The anomaly detection model was trained using several business performance indicators that represent financial, operational, and marketing activities.

Feature	Business Purpose
Revenue	Detect unusual sales values
Orders	Measure purchasing activity
Units	Detect abnormal product demand
Net Profit	Identify unexpected profitability changes
Net Margin	Monitor profit margin fluctuations
PPC Cost	Analyze advertising cost anomalies
PPC Sales	Evaluate paid advertising performance
Refund %	Detect abnormal refund behavior
Sessions	Measure customer traffic
Page Views	Analyze customer engagement
Clicks	Detect unusual advertising activity
Impressions	Monitor advertising exposure

4.5 Machine Learning Model

The project utilizes the Isolation Forest algorithm for anomaly detection.

Isolation Forest is an unsupervised machine learning algorithm specifically designed to identify abnormal observations within large datasets without requiring labeled training data.

Unlike supervised classification models, Isolation Forest isolates observations that differ significantly from the normal behavior of the majority of records. Transactions that are easier to isolate are more likely to be classified as anomalies.

The model was selected because it is highly efficient for large datasets, requires no predefined labels, and performs well in identifying rare or unexpected business events.

4.6 Model Output

After training, the model analyzes every transaction and assigns an anomaly prediction.

Each record is classified into one of the following categories:

Prediction	Description
Normal	The transaction follows expected business behavior.
Anomaly	The transaction significantly differs from normal business patterns and may require further investigation.

Additionally, the model generates an Anomaly Score for each record, representing the degree of abnormality. Higher scores indicate a greater likelihood that the transaction is unusual.

<https://github.com/DuaA-A/Enterprise-E-Commerce-Predictive-Analytics/tree/main/Ai%20Model>

4.7 Business Value

The integration of Machine Learning adds predictive intelligence to the project by automatically identifying unusual business behavior that may not be visible through traditional reporting.

The AI module provides several business benefits, including:

- Early detection of abnormal sales behavior.
- Identification of unexpected revenue and profitability fluctuations.
- Detection of unusual product performance.
- Reduction of manual data inspection.
- Faster investigation of operational issues.
- Improved decision-making through predictive analytics.

4.8 Summary

The Artificial Intelligence component extends the project beyond descriptive analytics by introducing automated anomaly detection. Using the Isolation Forest algorithm, the system identifies unusual business patterns without requiring labeled data.

This approach enables organizations to monitor operational performance proactively, detect potential business risks at an early stage, and support more effective data-driven decision-making.

Chapter 5

Dashboard and Visualization

5.1 Power BI Dashboard Pages

The Power BI solution consists of seven interactive analytical pages designed to provide a comprehensive view of the organization's performance. Each dashboard focuses on a specific business domain, enabling decision-makers to monitor financial performance, product profitability, marketplace efficiency, marketing effectiveness, quality control, promotional campaigns, and AI-driven predictive insights. The dashboards are connected through interactive navigation buttons and synchronized filters, allowing users to seamlessly explore business metrics from multiple perspectives.

Page	Title	Theme / Purpose
Home	Project Home	Landing page providing project introduction, navigation menu, and quick access to all dashboard pages.
1	Executive Overview	Presents overall business performance through executive KPIs, profitability trends, revenue analysis, and financial health indicators.
2	SKU Performance	Evaluates product-level performance, identifies top-selling products, analyzes margin erosion, and detects Bleeding SKUs using AI predictions.
3	Marketplace Performance	Compares business performance across different marketplaces by analyzing revenue, profitability, ROI, and geographic distribution.
4	Marketing & Traffic (Halo Effect)	Measures advertising performance through PPC analysis, traffic quality, conversion rates, and the Halo Effect between paid and organic sales.
5	Refund & Quality Control (R&Q Control)	Monitors refund behavior, financial losses, refund rate trends, high-risk SKUs, and product quality issues.
6	Promotions Analysis	Evaluates promotional campaign effectiveness through promo sales, promo ROI, promo conversion, and the relationship between discounts and refunds.
7	AI & Machine Learning	Presents predictive analytics results including Random Forest Bleeding SKU predictions, feature importance, and machine learning insights for decision support.

5.2 KPI Highlights

Across the six dashboard views, critical headline metrics are displayed via executive KPI cards:

Metric Displayed	Core Value / Status	Dashboard Location
Blended Net Margin %	Tracked vs. > 15% Target	Executive Command
Total TACoS %	Tracked vs. < 10% Benchmark	Executive Command
Bleeding SKU Count	Dynamic Alert Count	Product Portfolio
Best Performing Region	Comparative Marketplace ROI	Geographic Performance
PPC Conversion Rate	Ad Funnel Efficiency (%)	Marketing & Halo Effect
Total Refund Cost Impact (\$)	Absolute Dollar Leakage	Refund Leakage
Average Daily Velocity	Units Sold / Day	Promotion & Velocity

5.3 UI/UX Design Philosophy

To ensure rapid cognitive processing for corporate decision-makers, the dashboard layout adheres to strict data visualization principles:

- **Visual Hierarchy:** Top-level executive KPI cards sit at the upper fold, followed by macro-trend charts, and bottom-level granular data matrices.
- **Color Psychology:** Alert Red is reserved exclusively for negative ROI, Bleeding SKUs, and refund leakage. Growth Blue and Deep Teal signify healthy revenue growth and strong profit margins.
- **Dynamic Interactivity:** Slicers across all pages allow instant segmentation by Date Range, Marketplace (UK, DE, FR, IT, ES), Product Category, and SKU ID.

5.4 Figma Design

Before the implementation phase in Power BI, the entire dashboard interface was designed and prototyped using Figma. This design-first approach enabled the team to define the complete user experience, including page layouts, navigation flow, color palette, KPI placement, and visualization selection before development. As a result, the implementation process became more organized, reduced redesign efforts, and ensured visual consistency across all dashboard pages.

The Figma prototype served as the blueprint for the final Power BI solution, allowing every dashboard page to follow a unified storytelling approach while maintaining an executive-level user experience.

5.4.1 Visual Identity & Dashboard Design Philosophy

The dashboard was designed following a modern business intelligence philosophy that emphasizes simplicity, clarity, and executive decision-making. The primary objective was to transform complex e-commerce data into an intuitive analytical experience that enables users to identify trends, monitor business performance, and make data-driven decisions quickly.

Color Palette & Layout Design

A consistent green-and-white color palette was adopted throughout the dashboard to reflect business growth, financial performance, and data reliability. Soft neutral backgrounds were combined with green accent colors to improve readability while maintaining a clean corporate appearance.

Each dashboard page follows the same visual hierarchy:

- KPI cards positioned at the top for instant performance monitoring.
- Analytical charts placed in the center to visualize trends and comparisons.
- Supporting visuals located at the bottom to provide deeper business insights.
- A fixed navigation sidebar enabling users to move seamlessly between dashboard pages.

This standardized layout improves usability and creates a consistent experience across the entire dashboard.

Navigation & User Experience

A dedicated navigation panel was designed on the left side of every page to simplify dashboard exploration. Interactive navigation buttons allow users to switch instantly between analytical pages without returning to the home screen.

The dashboard also includes synchronized slicers and filters, enabling users to analyze business performance by:

- Date
- Marketplace
- Product (SKU)
- Brand

This interactive design allows executives to drill down into specific business areas while maintaining a smooth user experience.

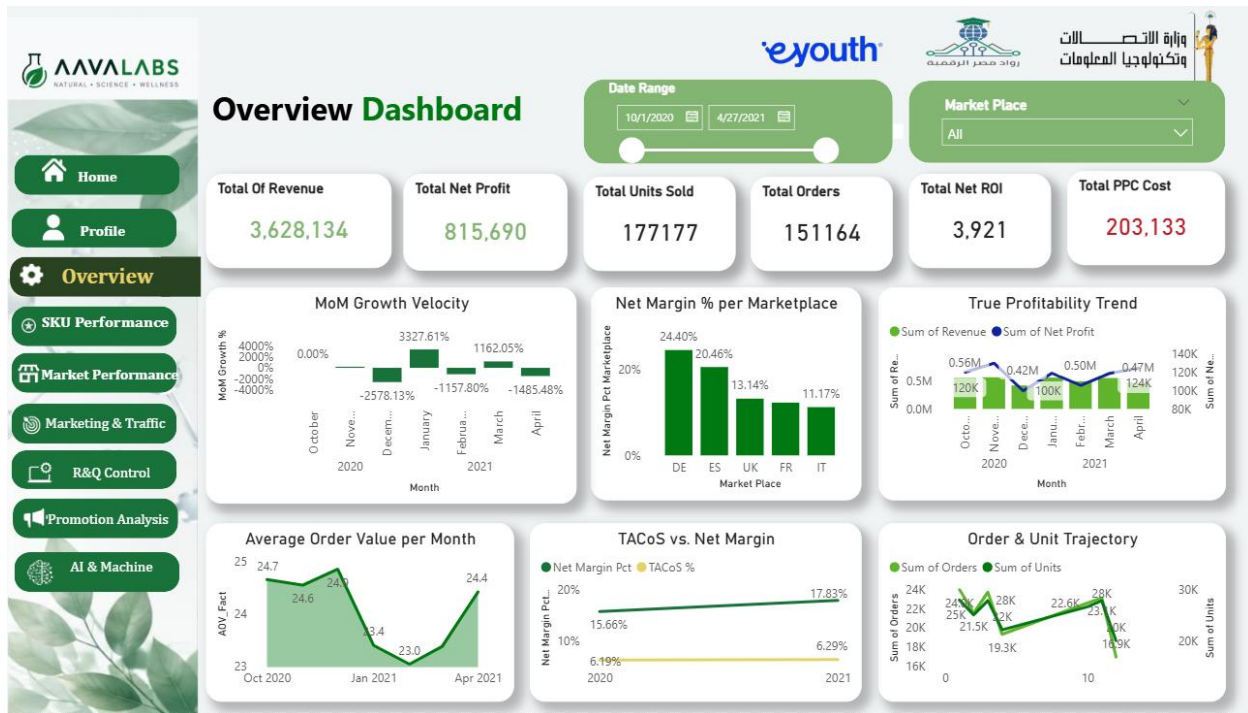
Dashboard Design Consistency

Each dashboard page was designed around a specific business objective while maintaining a consistent visual identity.

1. Executive Overview – Provides a high-level summary of revenue, profitability, operational performance, and business KPIs.
2. SKU Performance – Focuses on product-level analysis, top-performing products, margin erosion, and AI-based Bleeding SKU detection.
3. Marketplace Performance – Compares financial performance across different marketplaces and highlights regional profitability and operational efficiency.
4. Marketing & Traffic (Halo Effect) – Evaluates advertising performance, customer traffic, PPC campaigns, and the relationship between paid and organic sales.
5. Refund & Quality Control (R&Q Control) – Monitors refund behavior, quality issues, refund trends, and their financial impact on the business.
6. Promotions Analysis – Measures promotional campaign effectiveness, promotion ROI, promo conversion, and customer purchasing behavior during promotional periods.
7. AI & Machine Learning – Presents predictive analytics generated using the Random Forest model, including Bleeding SKU prediction, feature importance, and AI-driven business insights.

5.5 Dashboard Walkthrough

5.5.1 Executive Overview Dashboard



The Executive Overview dashboard provides a high-level summary of the company's overall business performance across all Amazon marketplaces. It is designed for decision-makers to quickly assess revenue, profitability, operational efficiency, and sales trends before exploring more detailed analytical pages.

Primary KPIs

- Total Revenue
- Total Net Profit
- Total Units Sold
- Total Orders
- Net ROI
- Total PPC Cost

Key Visualizations

- Monthly Revenue Growth
- Net Margin by Marketplace
- Profitability Trend Over Time

- Average Order Value Trend
- TACoS vs Net Margin Analysis
- Orders & Units Trend

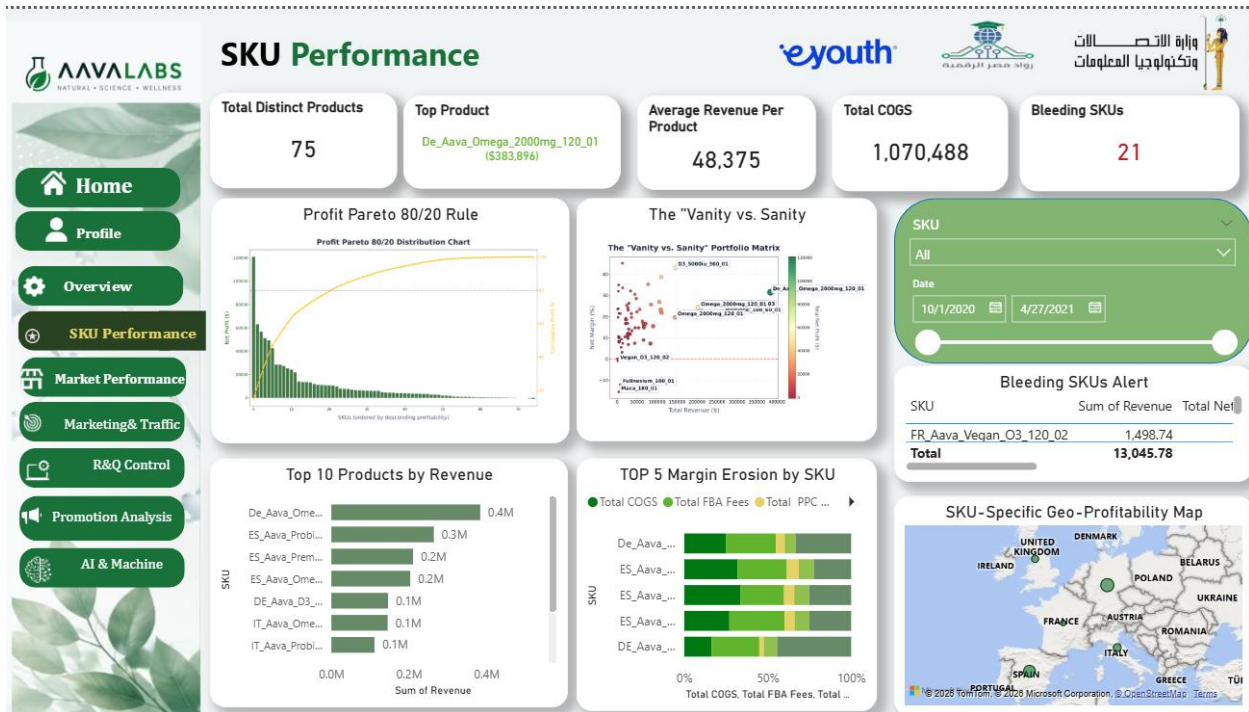
Filters

Filter	Purpose
Date	Filters all KPIs and visuals based on the selected time period.
Marketplace	Filters all metrics by the selected Amazon marketplace (UK, DE, FR, IT, ES).

Business Value

This dashboard enables executives to monitor overall business health, identify performance fluctuations, evaluate profitability across marketplaces, and make strategic decisions based on high-level KPIs.

5.5.2 SKU Performance Dashboard



The SKU Performance dashboard focuses on product-level analysis, helping identify top-performing products as well as underperforming SKUs that negatively impact profitability.

Primary KPIs

- Total Distinct Products
- Highest Product Revenue
- Average Revenue per Product
- Total COGS
- Number of Bleeding SKUs

Key Visualizations

- Top 10 Products by Revenue
- Bleeding SKU Alert Table
- SKU Profitability Distribution
- Profit Pareto Analysis (80/20)
- Geographic Profitability Map
- Margin Erosion by SKU

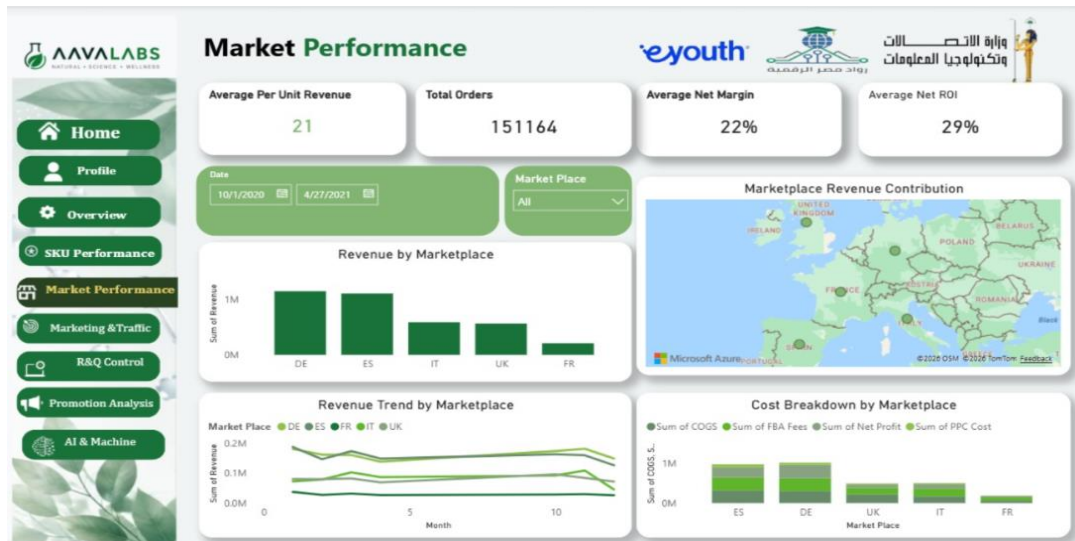
Filters

Filter	Purpose
Date	Displays SKU performance for the selected period.
SKU	Enables detailed analysis of an individual product (SKU).

Business Value

This page helps product managers identify profitable products, detect low-margin SKUs, optimize inventory allocation, and prioritize pricing or cost optimization strategies.

5.5.3 Marketplace & Geographic Performance Dashboard



This dashboard evaluates business performance across different European Amazon marketplaces, allowing management to compare regional sales, profitability, and operational efficiency.

Primary KPIs

- Revenue per Unit
- Total Orders
- Average Net Margin
- Average Net ROI

Key Visualizations

- Revenue by Marketplace
- Geographic Revenue Map
- Revenue Trend by Marketplace
- Marketplace Cost Breakdown

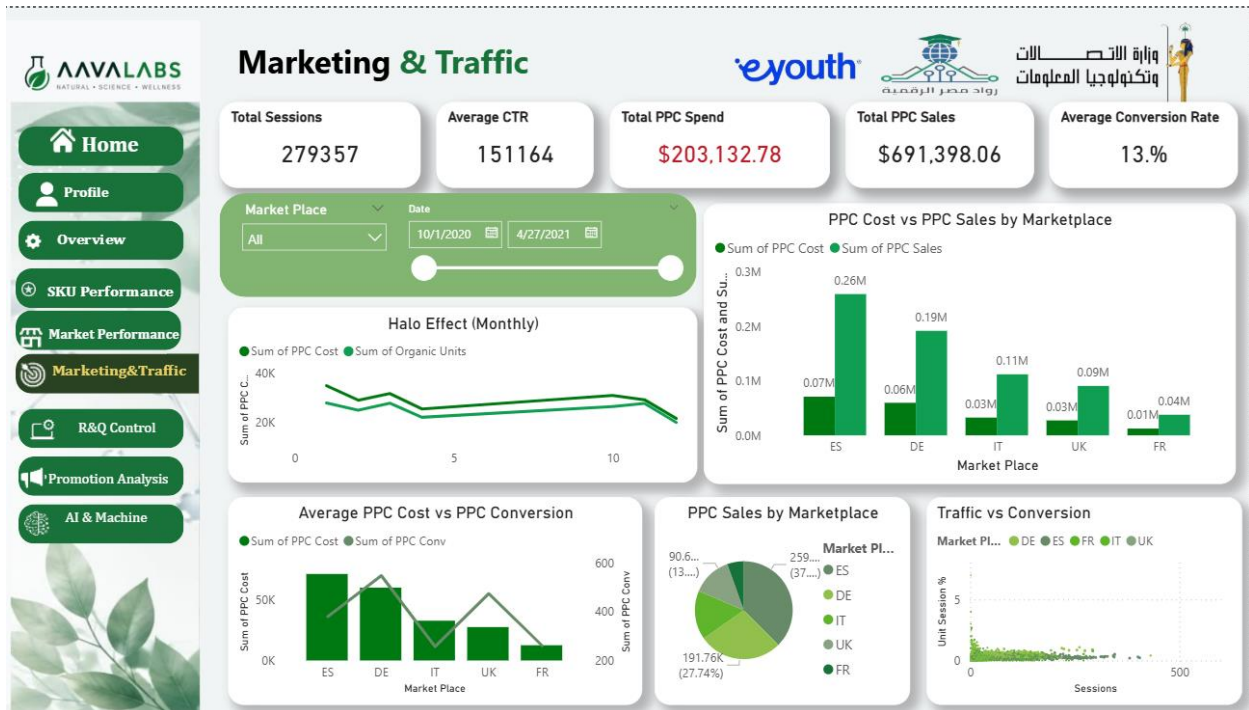
Filters

Filter	Purpose
Date	Filters marketplace performance by the selected period.
Marketplace	Allows comparison between Amazon marketplaces.

Business Value

The dashboard supports regional performance analysis, identifies the strongest and weakest marketplaces, and helps optimize expansion strategies across different countries.

5.5.4 Marketing, Traffic & Halo Effect Dashboard



This dashboard analyzes advertising performance by measuring the relationship between PPC campaigns and organic sales growth, enabling evaluation of the Halo Effect generated by paid advertising.

Primary KPIs

- Total Sessions
- Average CTR
- Total PPC Spend
- Total PPC Sales
- Average Conversion Rate

Key Visualizations

- Halo Effect Trend
- PPC Cost vs PPC Sales by Marketplace
- PPC Cost vs PPC Conversion
- Traffic vs Conversion Scatter Plot
- PPC Sales Distribution

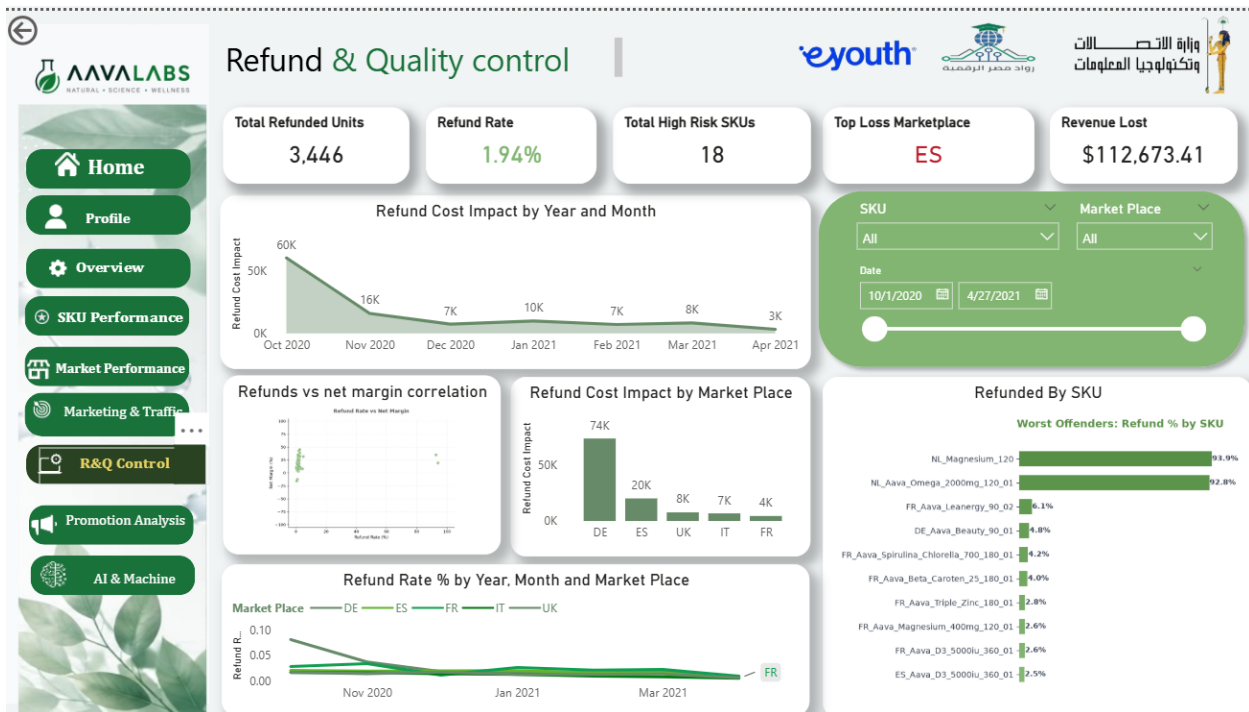
Filters

Filter	Purpose
Date	Filters marketing performance for the selected period.
Marketplace	Analyzes advertising metrics for a selected marketplace.

Business Value

This page helps evaluate marketing efficiency, optimize advertising budgets, measure campaign effectiveness, and understand how paid advertising influences organic product sales.

5.5.5 Refund Leakage & Quality Control Dashboard



The Refund Leakage dashboard focuses on product quality, refund behavior, and financial losses caused by customer returns.

Primary KPIs

- Total Refunded Units
- Refund Rate
- High-Risk SKUs
- Revenue Lost Due to Refunds
- Highest Refund Marketplace

Key Visualizations

- Refund Cost Trend
- Worst Offending SKUs
- Refund Rate vs Net Margin
- Refund Cost by Marketplace
- High-Risk Product Matrix

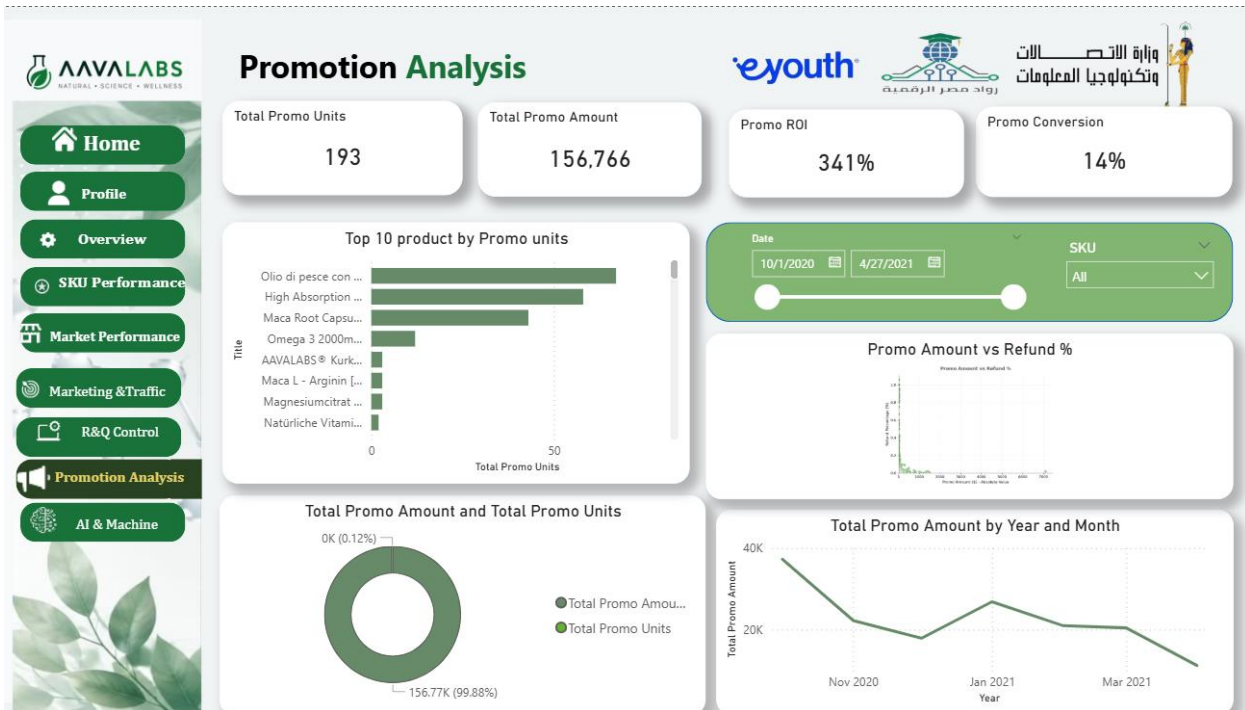
Filters

Filter	Purpose
SKU	Focuses the analysis on a specific product.
Marketplace	Displays refund and quality metrics for the selected marketplace.

Business Value

This dashboard helps reduce financial losses by identifying products with excessive refund rates, monitoring quality issues, and supporting operational improvements.

5.5.6 Promotions Analysis Dashboard



The Promotions Analysis dashboard evaluates the effectiveness of promotional campaigns by comparing promotional sales against organic sales while measuring their financial impact.

Primary KPIs

- Total Promo Units
- Total Promo Amount
- Promo ROI
- Promo Conversion Rate

Key Visualizations

- Organic vs Promotional Units
- Promo Amount Trend
- Top Products by Promo Units
- Promo Amount vs Refund Rate
- Promotion Performance Comparison

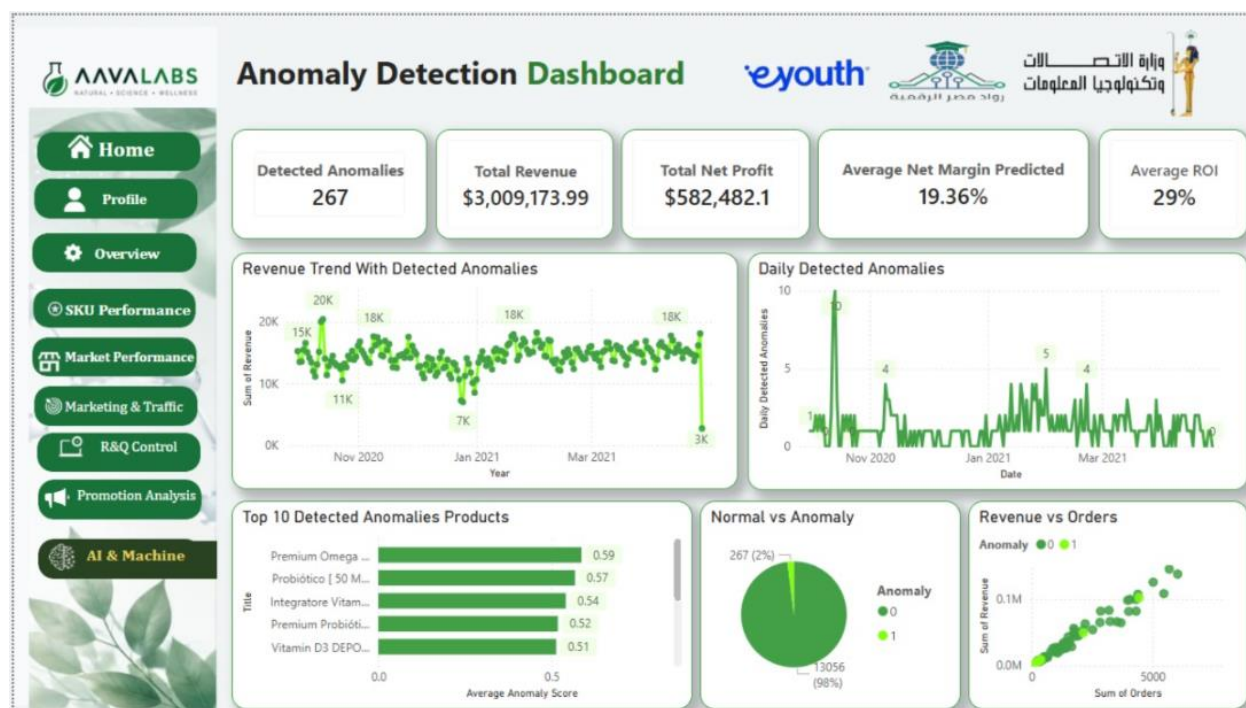
Filters

Filter	Purpose
Date	Filters promotional performance by the selected period.
SKU	Analyzes promotional impact for a specific product.

Business Value

This page enables marketing teams to evaluate campaign performance, determine promotion profitability, optimize discount strategies, and maximize promotional return on investment.

5.5.7 AI & Machine Learning Dashboard



The AI & Machine Learning dashboard focuses on detecting abnormal sales patterns, monitoring business anomalies, and supporting proactive decision-making through anomaly detection techniques. It enables stakeholders to identify unusual behaviors that may indicate operational issues, unexpected revenue fluctuations, or exceptional product **performance**.

Primary KPIs

- Detected Anomalies
- Total Revenue
- Total Net Profit
- Average Predicted Net Margin
- Average ROI

Key Visualizations

- Revenue Trend with Detected Anomalies
- Daily Detected Anomalies
- Top 10 Detected Anomalous Products
- Normal vs. Anomaly Distribution
- Revenue vs. Orders Scatter Plot

Business Value

This dashboard enables early identification of unusual sales behavior, supports proactive risk management, detects products with abnormal performance, and helps decision-makers investigate operational or financial anomalies before they significantly impact business performance.

5.6 Design Summary

The enterprise dashboard suite successfully transforms over 13,000 raw corporate records into an executive intelligence system. By integrating a clean SQL/Python Medallion data architecture with predictive Prophet and Random Forest machine learning outputs, the project delivers immediate commercial value:

Architecture Layer	Core Value Delivered	Strategic Business Outcome
Data Engineering	Reconciled 13,000+ records into a clean star schema.	Eliminates data silos and establishes 100% data trust across teams.
Machine Learning	60-day demand forecast + Bleeding SKU risk probabilities.	Prevents stockouts and proactively protects gross product margins.
BI Visualization	7 interactive Power BI command centers.	Empowers executives to optimize PPC budgets, cut logistics waste, and drive growth.

<https://github.com/DuaA-A/Enterprise-E-Commerce-Predictive-Analytics/tree/main/Power%20Bi>

Conclusion & Business Impact

The *Enterprise E-Commerce Intelligence & Predictive Analytics* project successfully bridges the critical gap between raw, unstructured transactional data and actionable business intelligence. By architecting a robust Medallion Data Pipeline, we transformed over 13,000 disparate records into a centralized, highly optimized Star Schema Data Warehouse.

This infrastructure empowered the deployment of advanced Exploratory Data Analysis (EDA) and Predictive Machine Learning models, effectively shifting the business posture from reactive observation to proactive strategy. Through the deployment of six specialized Power BI dashboards, the organization now possesses granular visibility into critical operational metrics—ranging from identifying "Bleeding SKUs" and refund leakage to optimizing promotional ROI and predicting future demand.

Ultimately, this project delivers a scalable, automated analytics engine that actively guards profit margins, enhances inventory efficiency, and provides stakeholders with the empirical foundation required for sustainable, data-driven e-commerce growth.